Handover: Heart Failure Admissions Reporting Pipeline

This handover document outlines the end-to-end process for regenerating the Heart Failure Admissions Executive Summary. The report is designed with a "frozen data" architecture, separating data processing from the final visual rendering to ensure reproducibility and auditability.

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1. Overview

This project generates a professional 3-page PDF executive summary comparing Heart Failure (HF) admissions to non-HF admissions using the MIMIC-III database. It tracks demographics, clinical outcomes (LOS, mortality), and prescription patterns.

2. System Requirements & Setup

To regenerate this report, the following must be installed:

- R & RStudio: Version 4.1 or higher.
- Quarto CLI: For document rendering.
- TeX Distribution: TinyTeX is required for PDF output. If missing, run quarto install tinytex in the terminal.
- R Packages:

  R

  install.packages(c("tidyverse", "knitr", "gt", "gtsummary", "DBI", "readr"))

3. Data Source & Processing (Step 1)

The first stage involves two main steps, first generating SQL tables featuring the necessary data, and then pulling this data into R for advanced analysis and visualisation.

- Source Data: The R script pulls from two SQL views/tables: public.hf_cohort and public.hf_drug_summary_view.
- Key Logic:
  - Cohort Creation: Segments patients based on ICD-9 codes (428.x, and 429.x).
  - Table Formatting: Uses gtsummary for demographics and gt for outcomes and medications.
- Crucial Step: You must ensure the SQL connection object conn is active before running.
- Output: The script generates four "frozen" data objects saved as .rds files. These files must be saved in the same folder as the Quarto (.qmd) file:

  1. q1_hf_outcomes_summary.rds (HF Demographics)
  2. q1_non_hf_outcomes_summary.rds (Non-HF Demographics)
  3. q2_outcomes_summary.rds (Clinical Outcomes)
  4. q3_prescriptions_summary.rds (Medication Patterns)

4. Report Rendering (Step 2)

The final report is generated by rendering the Heart-Failure-Report.qmd file.

- Logic: The Quarto file does not connect to the database. Instead, it uses readRDS() to load the four files generated in Step 1.
- Execution: Open the .qmd file in RStudio and click the Render button (or run quarto render Heart-Failure-Report.qmd --to pdf in the terminal).

5. Critical Formatting Notes (Lessons Learned)

To maintain the professional look and keep the report under the 3-page limit, specific formatting was applied that must be preserved:

- The as_gt() Bridge: Demographics created with tbl_summary must be converted using as_gt() before applying gt styling.
- Space Management: To prevent the report from exceeding 3 pages, row padding is set to px(1) and font sizes are set to px(10) or px(9) within the tab_options() function of the gt tables.
- Column Widths: The outcomes table uses cols_width() to prevent text wrapping, which otherwise adds unnecessary vertical height.

6. Troubleshooting

- "Cannot coerce class 'function' to a data.frame": This error occurs if an RDS file is missing from the folder. R defaults to a built-in function of the same name. Ensure all .rds files are present.
- Table Bleeding: If tables go off the side of the page, ensure table.width = pct(100) is applied in the gt options.
- LaTeX Errors: If the PDF fails to render, check that library(gt) and library(gtsummary) are loaded in the Quarto setup chunk, as they are required to interpret the RDS objects.

7. Maintenance & Updates

For future reporting cycles (e.g., next year), update the SQL views in MIMIC-III, run the analysis script once to refresh the .rds files, and re-render the Quarto document. No changes to the Quarto text should be necessary unless the report objectives change.

Appendix

Within the code pipeline, there are 7 scripts which should be run in the following order:

1. SQL script (pulls the data and creates tables in SQL)
2. Load data.R (connects R to SQL and loads the data)
3. Heart failure EDA.R (performs EDA on the HF tables)
4. Heart failure final products.R (Generates near ready tables for Quarto and saves RDS objects in memory)
5. Prescription EDA.R (Does EDA for theprescription dataset)
6. Prescription final products.R (Generates the prescription tables for Quarto)
7. Heart failure Report. Qmd (Generates the Quarto report)